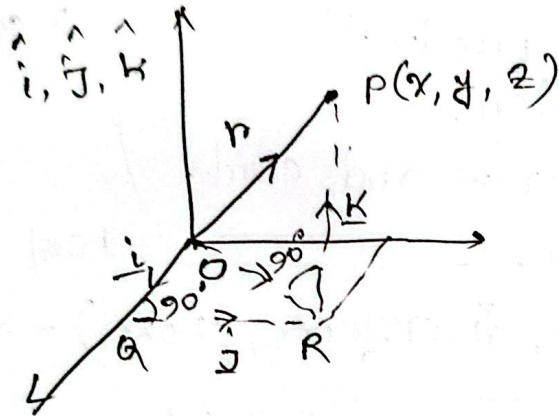


Vector: A quantity which has both magnitude and direction is called vector.

Position vector:



Let O be the origin of the coordinate system. Then the vector $\vec{r} = \vec{OP}$ is called the position vector of the point $P(x, y, z)$

$$\begin{aligned} OQ &= x, & \vec{OQ} &= x\hat{i} \\ OR &= y, & \vec{OR} &= y\hat{j} \\ RP &= z, & \vec{RP} &= z\hat{k} \end{aligned}$$

$$\begin{aligned} OP &= OR + RP \\ &= OQ + QR + RP \\ &= x^2 + y^2 + z^2 \end{aligned}$$

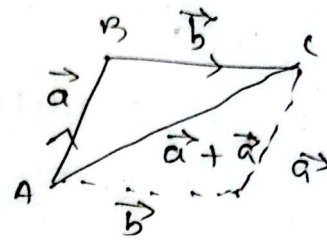
$$OP = \sqrt{x^2 + y^2 + z^2}$$

$$\begin{aligned} \vec{OP} &= \vec{r} = \vec{OR} + \vec{RP} \\ &= \vec{OQ} + \vec{QR} + \vec{RP} \\ &= x\hat{i} + y\hat{j} + z\hat{k} \end{aligned}$$

* Laws:-

i) commutative Law:-

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$



From the triangle $\triangle ABC$,

$$\vec{AB} + \vec{BC} = \vec{AC}$$

$$\Rightarrow \vec{a} + \vec{b} = \vec{AC} \quad \dots (i)$$

From the triangle $\triangle ADC$,

$$\vec{AD} + \vec{DC} = \vec{AC}$$

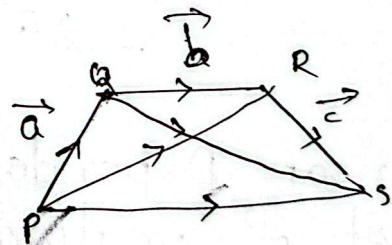
$$\Rightarrow \vec{b} + \vec{a} = \vec{AC} \quad \dots (ii)$$

$\therefore$ Vectors always follow the commutative law.

ii) Associative Law:-

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

$$\begin{aligned} (\vec{a} + \vec{b}) + \vec{c} &= (\vec{PQ} + \vec{QR}) + \vec{RS} \\ &= \vec{PR} + \vec{RS} \\ &= \vec{PS} \end{aligned}$$

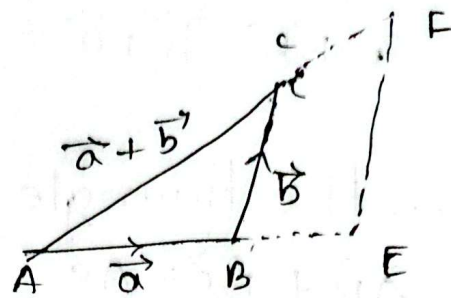


$$\begin{aligned} \vec{a} + (\vec{b} + \vec{c}) &= \vec{PQ} + (\vec{QR} + \vec{RS}) \\ &= \vec{PQ} + \vec{QS} \\ &= \vec{PS} \end{aligned}$$

(iii) Distributive law:

where, $\lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}$
 λ is a scalar

Let,
 $\vec{AE} = \lambda\vec{AB}$



$\triangle ABC$ and $\triangle AEF$ is a similar triangle.

$$\therefore \frac{AE}{AB} = \frac{EF}{BC} = \frac{AF}{AC} = \lambda \text{ (say)}$$

$$\therefore \vec{AE} = \lambda\vec{AB}, \vec{EF} = \lambda\vec{BC}, \vec{AF} = \lambda\vec{AC}$$

$$\therefore \vec{AE} = \lambda\vec{a}, \vec{EF} = \lambda\vec{b}, \vec{AF} = \lambda(\vec{a} + \vec{b})$$

From, $\triangle AEF$,

$$\begin{aligned} \vec{AE} &= \vec{AF} + \vec{EF} \\ &= \lambda\vec{a} + \lambda\vec{b} \end{aligned}$$

* Linear Dependence and Independence of a set of vectors:-

Definition:- If there exists a set of vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ and the equal number of scalars $a_1, a_2, \dots, a_n$. Then the sum of the product,

$$a_1\vec{a}_1 + a_2\vec{a}_2 + \dots + a_n\vec{a}_n \quad (1)$$

is called the linear combination of the

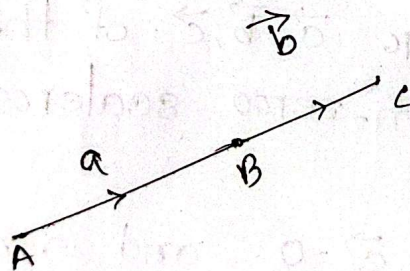
vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$.

Trivial case:

i) If all the scalars, $(\alpha_1, \alpha_2, \dots, \alpha_n)$ are ^{not} equal to zero. Then But the linear combination $\dots$ (i) is equal to zero, then the vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ are linearly dependent.

ii) If the scalars $(\alpha_1, \alpha_2, \dots, \alpha_n)$ are all zero to make the linear combination $\dots$ (i) equal to zero, then vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots, \vec{a}_n$ are linearly independent.

From Geometrical point of view:-



$$\vec{AB} = \vec{a}$$

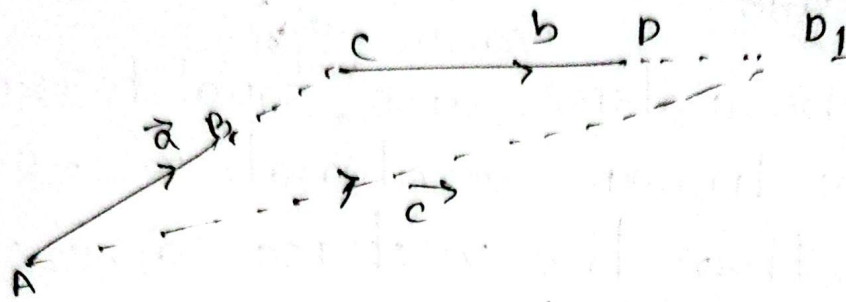
$$\vec{AC} = \vec{b}$$

$$\vec{AC} = \lambda \vec{AB}$$

$$\Rightarrow \vec{b} = \lambda \vec{a}$$

$\Rightarrow$ Linearly dependent (co-linear).

two vectors in a plane:



$$\overrightarrow{AB} = \vec{a}$$

$$\overrightarrow{AD} = \vec{b}$$

$$\overrightarrow{AC} = \lambda \vec{a}$$

$$\overrightarrow{CD} = \lambda_1 \vec{b}$$

$$\vec{c} = \lambda \vec{a} + \lambda_1 \vec{b}$$

gf. two vectors are in same plane, then the vectors are linearly dependent. (coplanar vector).

* gf the position vectors $\vec{a}, \vec{b}, \vec{c}$ of the three points A, B, C and the non-zero scalars α, β, γ are such that,

$$\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c} = 0 \quad \text{and} \quad \alpha + \beta + \gamma = 0$$

then the terminal points of the three vectors are collinear.

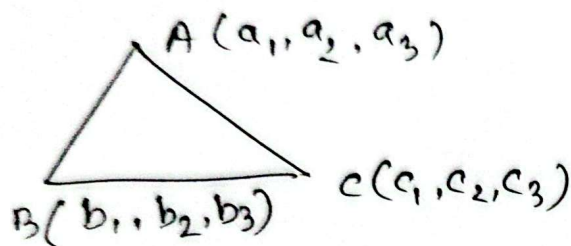
* gf the position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ of four points A, B, C, D and the non-zero scalars $\alpha, \beta, \gamma, \delta$ are such that $\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c} + \delta \vec{d} = 0$ and $\alpha + \beta + \gamma + \delta = 0$, then the four points are coplanar.

$$\vec{a} = 1\hat{i} - 3\hat{j} + 5\hat{k}$$

$$\vec{b} = \dots$$

$$\vec{c} = \dots$$

Prove that



$$\therefore \vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\vec{B} = \dots$$

$$\vec{C} = \dots$$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0 \rightarrow \text{Linearly dependant} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{Applicable} \\ \text{for} \\ \text{three} \\ \text{vector.} \end{array}$$

$$\neq 0 \rightarrow \text{Linearly Independent}$$

Prob. from AK. Hazra:-

4. Show that the three vectors $\vec{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} - 4\hat{k}$, $\vec{C} = 4\hat{i} + 3\hat{j} - \hat{k}$ are linearly dependent and determine a relation between them and hence show that the terminal points are collinear.

$$\begin{vmatrix} 2 & 1 & -3 \\ 1 & 0 & -4 \\ 4 & 3 & -1 \end{vmatrix} = 0$$

$\therefore$, three vectors are linearly dependent.

Relation:-

$$\alpha \vec{A} + \beta \vec{B} + \gamma \vec{C} = 0 \quad \dots (i) \Rightarrow$$

$$\Rightarrow \alpha(2\hat{i} + \hat{j} - 3\hat{k}) + \beta(\hat{i} - 4\hat{k}) + \gamma(4\hat{i} + 3\hat{j} - \hat{k}) = 0 \quad \dots (ii)$$

$$\therefore 2\alpha + \beta + 4\gamma = 0 \quad \dots (i)$$

$$\alpha + 0\beta + 3\gamma = 0 \quad \dots (ii)$$

$$-3\alpha - 4\beta - \gamma = 0 \quad \dots (iii)$$

(i) Unit Solution

(ii) Infinite many solution

from equation (i) and (ii)

$$\frac{\alpha}{3} = \frac{\beta}{-2} = \frac{\gamma}{-1} = k \quad (\text{say})$$

$$\alpha = 3k$$

$$\beta = -2k$$

$$\gamma = -k$$

putting the values in equation (i) $\Rightarrow$

$$\alpha \vec{A} + \beta \vec{B} + \gamma \vec{C} = 0$$

$$\Rightarrow 3k\vec{A} - 2k\vec{B} - k\vec{C} = 0$$

$$\Rightarrow 3\vec{A} - 2\vec{B} - \vec{C} = 0$$

Ans.

And, $\alpha + \beta + \gamma = 0 \therefore$ terminal points are collinear.

* Determine whether the vectors $\vec{r}_1 = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{r}_3 = -2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{r}_4 = 3\hat{i} + 2\hat{j} + 5\hat{k}$ are linearly dependent or not,

$$\alpha_1 \vec{a}_1 + \alpha_2 \vec{a}_2 + \alpha_3 \vec{a}_3 + \alpha_4 \vec{a}_4 = 0$$

$$\Rightarrow \alpha_4 \vec{a}_4 = -(\alpha_1 \vec{a}_1 + \alpha_2 \vec{a}_2 + \alpha_3 \vec{a}_3)$$

$$\Rightarrow \vec{a}_4 = -\left(\frac{\alpha_1}{\alpha_4} \vec{a}_1 + \frac{\alpha_2}{\alpha_4} \vec{a}_2 + \frac{\alpha_3}{\alpha_4} \vec{a}_3\right)$$

$$\Rightarrow \vec{a}_4 = \lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \lambda_3 \vec{a}_3$$

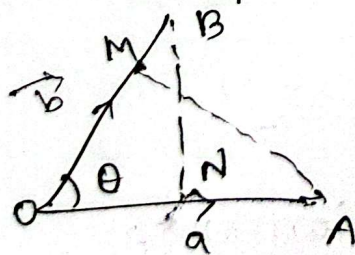
2nd Chapter

The dot and cross product

* The dot product of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{a} \cdot \vec{b}$ and is defined by,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = ab \cos \theta$$

Geometrically:-



$$\boxed{\text{proj}_{\vec{a}} \vec{b} = \vec{b} \cdot \hat{a}}$$

$$\therefore \vec{a} \cdot \vec{b} = ab \cos \theta$$

$$= a(b \cos \theta) = |\vec{a}| \cdot ON$$

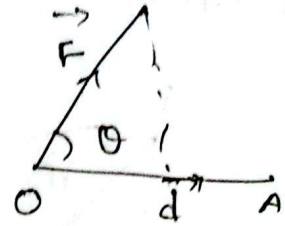
$$= b(a \cos \theta) = |\vec{b}| \cdot \text{The projected length of vector } \vec{b} \text{ upon vector } \vec{a}$$

$$= |\vec{b}| \cdot \text{proj}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \hat{a} \cdot \vec{b} = \vec{b} \cdot \hat{a}$$

Physically:-

Work done:-

The work done by the force $\vec{F}$ is defined by the product of the projected length of the force $\vec{F}$ in the direction of the displacement and the magnitude of the displacement.

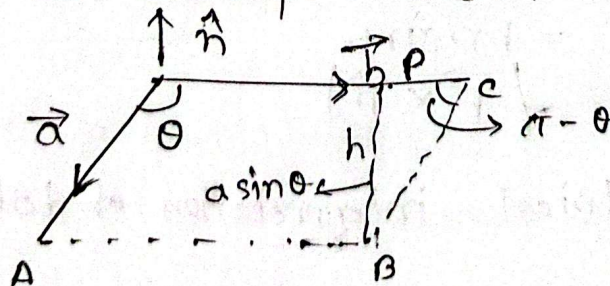


$$\begin{aligned}\therefore W &= (OA) \times |\vec{d}| \\ &= F \cos \theta \cdot d \\ &= Fd \cos \theta.\end{aligned}$$

Cross Product:- The cross product of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{a} \times \vec{b}$ and is defined by

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

Geometrical Interpretation:-



Area of parallelogram:-

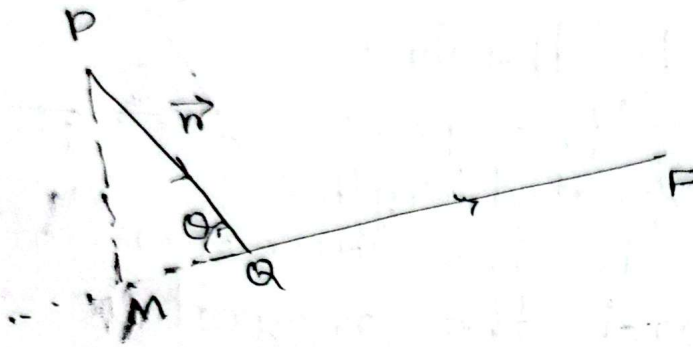
$$= (\text{Base}) \times \text{height}$$

$$= (\vec{b}) \cdot PB$$

$$= \vec{b} \cdot h = ab \sin \theta = |\vec{a} \times \vec{b}|$$

$$\therefore |\vec{a} \times \vec{b}| = ab \sin \theta$$

* physical interpretation:-
A cross product:-



Spiege problem No:- 35 :-

Find the expression for the moment of a force $\vec{F}$ about a point P.

Soln:-

The moment of a force $\vec{F}$ is defined as the product of the magnitude of the force $\vec{F}$ and perpendicular distance from a point to the line of action of $\vec{F}$.

$$\therefore \vec{M} \text{ moment} = |\vec{F}| \cdot r \sin \theta$$

$$= Fr \sin \theta$$

$$= |\vec{F} \times \vec{r}|$$

* Q-13 $\longleftrightarrow$ Geometrical interpretation of dot product

Ex:- 63, 64, 65, 67, 89

Physic - - - dot product

Q-17

Ex:- 69.

∞



∞ Geomet - - - - cross product :-

$Q \rightarrow 30, 32,$

$Ex \rightarrow 82, 84,$

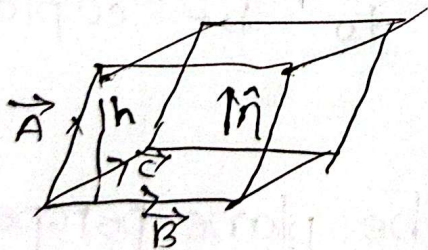
∞ Physical - - - - cross product :

$Q \rightarrow$

$Ex \rightarrow 86$

Triple product : $Q \rightarrow 37, Ex :- 43, 92$

Q-37:- Show that $\vec{A} \cdot (\vec{B} \times \vec{C})$ is an absolute value equal to the volume of a parallelepiped with edges are the vectors $\vec{A}, \vec{B}, \vec{C}$.



$$\vec{A} \cdot \vec{A} = A \cos \theta$$

Volume of a parallelepiped

= Area of the base $\times$ height

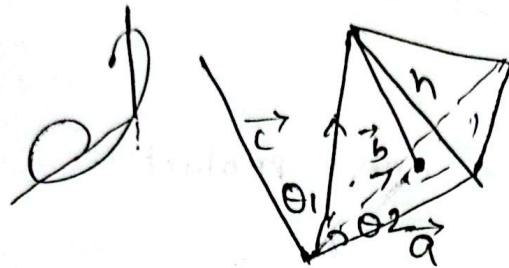
$$= |\vec{B} \times \vec{C}| \cdot h$$

$$= |\vec{B} \times \vec{C}| A \cos \theta$$

$$= \vec{A} \cdot \hat{n} |\vec{B} \times \vec{C}|$$

$$= \vec{A} \cdot (\vec{B} \times \vec{C})$$

* Volume of a tetrahedron:



$$\begin{aligned}
 \text{Volume} &= \frac{1}{3} (\text{Area of the base}) \times \text{height} \\
 &= \frac{1}{3} \left| \frac{1}{2} \vec{a} \times \vec{b} \right| \cdot c \cos \theta_1 \\
 &= \frac{1}{6} \vec{a} \cdot \vec{b} \times \vec{c} \cdot \sin \theta_2 \cdot \cos \theta_1 \\
 &= \frac{1}{6} \vec{a} \cdot (\vec{b} \times \vec{c}) \text{ or } \frac{1}{6} \vec{b} \cdot (\vec{a} \times \vec{c}) \text{ or } \frac{1}{6} \vec{c} \cdot (\vec{a} \times \vec{b})
 \end{aligned}$$

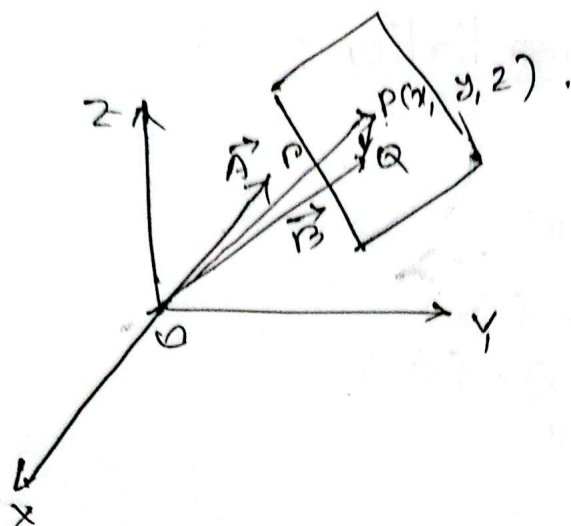
* Equation of the plane:-

Q-18, 19

Ex-77.

43. prove that the necessary and sufficient condition to the $\vec{A}, \vec{B}, \vec{C}$ to be coplanar is that $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

18-19: Find an equation of the plane perpendicular to the vector $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and passing through the terminal point of the vector $\vec{B} = \hat{i} + 5\hat{j} + 3\hat{k}$



The position vectors of the point $p(x, y, z)$ of the plane is :-

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{OP} + \vec{PQ} = \vec{OQ}$$

$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= \vec{B} - \vec{r}$$

$$= (1-x)\hat{i} + (5-y)\hat{j} + (3-z)\hat{k}$$

$\therefore$ The equation of the plane $= \vec{PQ} \cdot \vec{A} = 0$

$$\Rightarrow (1-x) \cdot 2 + (5-y) \cdot 3 + 6(3-z) = 0$$

$$\Rightarrow 2 - 2x + 15 - 3y + 18 - 6z = 0$$

$$\Rightarrow 2x + 3y + 6z = 35$$

Q19. Find the distance from the origin to the plane :

$$\text{distance} = \text{Pro}_{\vec{a}} \vec{b} = \vec{b} \cdot \hat{a}$$

$$\text{or, distance} = \frac{2 \cdot 0 + 3 \cdot 0 + 6 \cdot 0 - 35}{\sqrt{2^2 + 3^2 + 6^2}}$$

$$= 5$$

* Vector Differentiation:

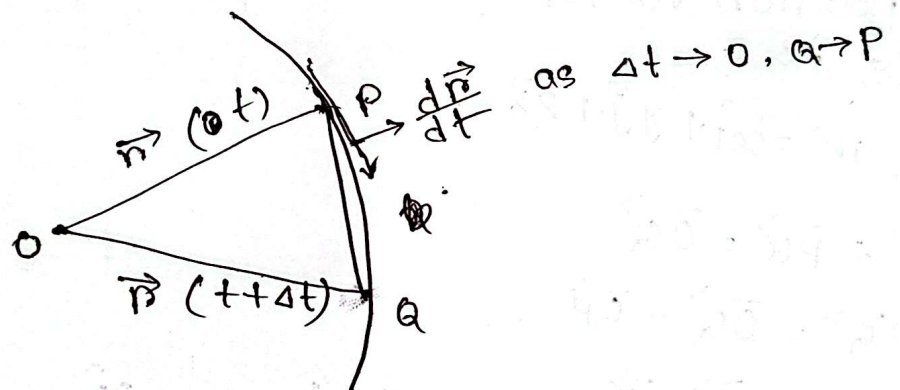
$$y = f(x)$$

$$y + \Delta y = f(x + \Delta x)$$

$$\Delta y = f(x + \Delta x) - f(x)$$

$$\Rightarrow \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \dots$$



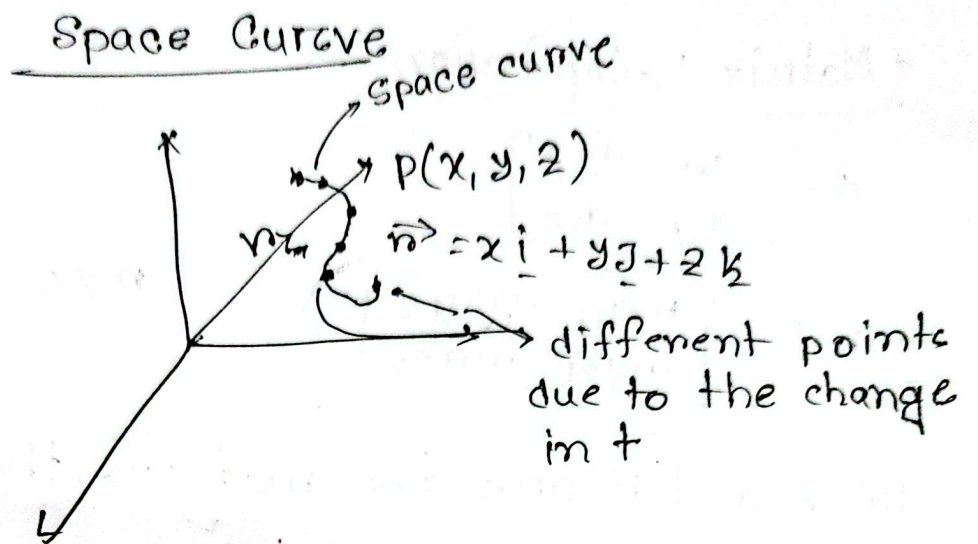
$$\frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$$

$$\Rightarrow \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$$

$$\Rightarrow \frac{d\vec{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$$

direction towards the tangent at p.

Space Curve



$$x = x(t)$$

$$y = y(t)$$

$$z = z(t)$$

$$\therefore \vec{r} = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}$$

$$\Rightarrow d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

* Find the unit tangent ^{vector} of the space curve point $t=2$ of the curve $x=t^2+1$, $y=4t-3$, $z=2t^2-6t$.

Solution:-

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$= (t^2+1)\vec{i} + (4t-3)\vec{j} + (2t^2-6t)\vec{k}$$

$$\therefore \frac{d\vec{r}}{dt} = \dots$$

$$\text{Unit tangent vector } \vec{T} = \frac{\frac{d\vec{r}}{dt}}{|\frac{d\vec{r}}{dt}|}$$

* Find the angle between the space tangent of the space curve at $t=\pm 2$.

$$\ast \frac{d\vec{r}}{dt} \Rightarrow \text{velocity } \vec{v}$$

$$\ast \text{acceleration } \vec{a} = \frac{d\vec{v}}{dt}$$

* A particle moves along the space curve $x=2t^2$, $y=t^2-4t$, $z=2t-5$, where t is the time. Find the components of its velocity and acceleration at time $t=1$, in the direction $\vec{i} - 3\vec{j} + 2\vec{k}$.

Solution:-

$$\vec{v} = \frac{d\vec{r}}{dt} = \text{Ans.}$$

$$\text{acceleration, } \vec{a} = \frac{d\vec{v}}{dt} = \text{Ans.}$$

third part:-

Velocity component in the given direction,

$$\text{proj } \frac{\vec{b}}{\vec{a}} = \vec{b} \cdot \vec{a} = \vec{v} \cdot \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}$$

$$\text{acceleration component} = \vec{a} \cdot \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}$$

Spice

9. If vector $\vec{A}$ has constant magnitude then show that $\vec{A}$ and $\frac{d\vec{A}}{dt}$ are perpendicular to each other.

Soln:-

$$\vec{A} \cdot \vec{A} = A^2 = \text{constant}$$

$$\Rightarrow \frac{d}{dt} (\vec{A} \cdot \vec{A}) = 0$$

$$\Rightarrow \vec{A} \cdot \frac{d\vec{A}}{dt} + \frac{d\vec{A}}{dt} \cdot \vec{A} = 0$$

$$\Rightarrow \vec{A} \cdot \frac{d\vec{A}}{dt} = 0$$

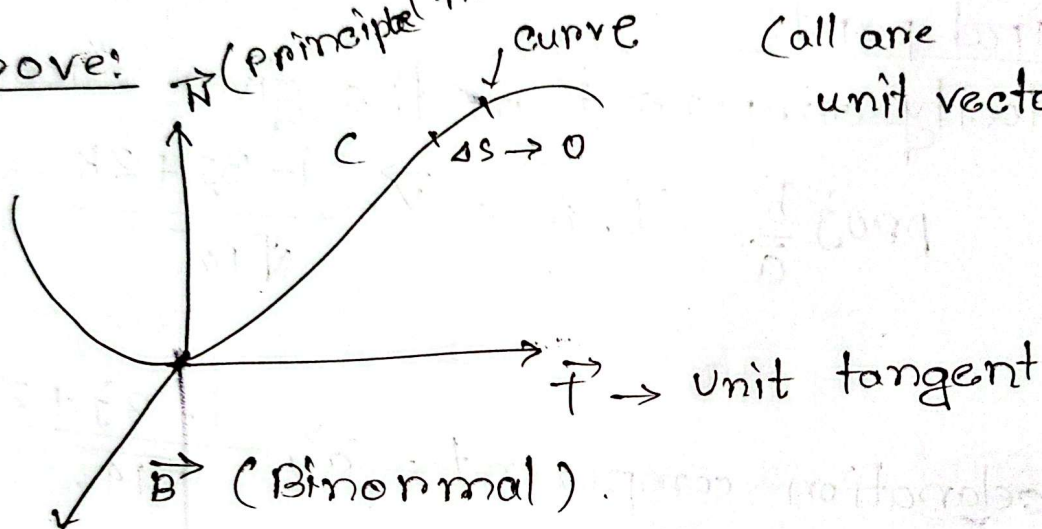
18. Frenet-Serret formula:-

$$(i) \frac{d\vec{T}}{ds} = k\vec{N}$$

$$(ii) \frac{d\vec{B}}{ds} = -\tau\vec{N}$$

$$(iii) \frac{d\vec{N}}{ds} = \tau\vec{B} - k\vec{T}$$

(iv) prove: $\vec{N}$ (principal normal)



(all are unit vectors)

$$\vec{T} \cdot \vec{T} = 1$$

$$\Rightarrow \vec{T} \times \frac{d\vec{T}}{ds} + \frac{d\vec{T}}{ds} \times \vec{T} = 0$$

$$\Rightarrow 2 \cdot \vec{T} \times \frac{d\vec{T}}{ds} = 0$$

$$\Rightarrow \vec{T} \times \frac{d\vec{T}}{ds} = 0$$

$\therefore \vec{N}$ and $\frac{d\vec{T}}{ds}$ are parallel,

$\therefore$ we can write:-

$$\frac{d\vec{T}}{ds} = k\vec{N}$$

where k is the ^{curvature} ~~curvature~~ and $\rho = \frac{1}{k}$ is the radius of curvature.

$$(ii) \quad \vec{B} \cdot \vec{B} = 1$$

$$\therefore \vec{B} \cdot \frac{d\vec{B}}{ds} = 0$$

$$\text{Let, } \vec{T} \times \vec{N} = \vec{B}$$

$$\Rightarrow \frac{d\vec{B}}{ds} = \vec{T} \times \frac{d\vec{N}}{ds} + \frac{d\vec{T}}{ds} \times \vec{N}$$

$$= \vec{T} \cdot \frac{d\vec{N}}{ds} + \kappa \vec{N} \times \vec{N} \rightarrow 0$$

$$\Rightarrow \vec{T} \cdot \frac{d\vec{B}}{ds} = \vec{T} \cdot (\vec{T} \times \frac{d\vec{N}}{ds})$$

spgex 42: $\vec{A} \cdot (\vec{A} \times \vec{C}) = 0$

$$\therefore \vec{T} \cdot \frac{d\vec{B}}{ds} = 0$$

$$\therefore \frac{d\vec{B}}{ds} \text{ is } \text{per parallel with } \vec{N}$$

$$\therefore \frac{d\vec{B}}{ds} = \tau \vec{N}$$

where τ is the torsion and $\sigma = \frac{1}{\tau}$ is the radius of torsion.

* 13 types of questions can be made from this
(iii) three equations.

(iii)

Let,

$$\vec{B} \times \vec{T} = \vec{N}$$

$$\begin{aligned}\Rightarrow \frac{d\vec{N}}{ds} &= \vec{B} \times \frac{d\vec{T}}{ds} + \frac{d\vec{B}}{ds} \times \vec{T} \\ &= \vec{B} \times k\vec{N} + (-\tau\vec{N} \times \vec{T}) \\ &= -k\vec{T} + \tau\vec{B} \\ &= \tau\vec{B} - k\vec{T}\end{aligned}$$

19. Given the space curve $x = 3\cos t$, $y = 3\sin t$, $z = 4t$, where t is the time. Find $\vec{T}$, $\vec{N}$, $\vec{B}$, $\vec{K}$, ρ , τ , ϕ .

Soln:-

$$\vec{T} = \frac{d\vec{r}/dt}{|d\vec{r}/dt|}$$

$$\vec{r} = 3\cos t \underline{i} + 3\sin t \underline{j} + 4t \underline{k}$$

$$\therefore \frac{d\vec{r}}{dt} = -3\sin t \underline{i} + 3\cos t \underline{j} + 4 \underline{k}$$

$$\therefore d\vec{T} = \frac{-3\sin t \underline{i} + 3\cos t \underline{j} + 4 \underline{k}}{5}$$

We know that,

$$\frac{d\vec{T}}{ds} = k\vec{N}$$
$$\Rightarrow \frac{\frac{d\vec{T}}{dt}}{\frac{ds}{dt}} = k\vec{N} \quad \dots \dots (1)$$

$$\therefore \frac{d\vec{T}}{dt} = -\frac{3}{5} \cos t \vec{i} - \frac{3}{5} \sin t \vec{j}$$

$$\text{and, } \left| \frac{d\vec{T}}{dt} \right| = \frac{ds}{dt} = 5$$

$\therefore (i) \Rightarrow$

$$-\frac{3}{5} \cos t \vec{i} - \frac{3}{5} \sin t \vec{j} = k\vec{N} \quad \dots \dots (ii)$$

$$\therefore k = \frac{3}{5}$$

$$\therefore \vec{N} = \frac{5}{3} \times \left(-\frac{3}{5} \cos t \vec{i} - \frac{3}{5} \sin t \vec{j} \right)$$

problem -22:

$$x=t, \quad y=t^2, \quad z=\frac{2}{3}t^3$$

Exercise-47:

$$t = 3 + \frac{1}{t^2}$$

Q-23 :- Find the equations in vector and rectangular form for the (a) tangent, (b) principal normal and (c) the binormal to the curve of Prob: 22 at the point where $t=1$.

Q-22 :- Given the space curve $x=t, y=t^2, z=\frac{2t^3}{3}$. Find the curvature κ and torsion τ .

Soln :-

Equation of the tangent :-

$$(\vec{r} - \vec{r}_0) \times \vec{T}_0 = 0 \quad \dots (i)$$

From soln of 22 :-

$$\vec{T} = \frac{\hat{i} + 2t\hat{j} + 2t^2\hat{k}}{1 + 2t^2}$$

Equation of the Normal

$$(\vec{r} - \vec{r}_0) \times \vec{N}_0 = 0 \quad \dots (ii)$$

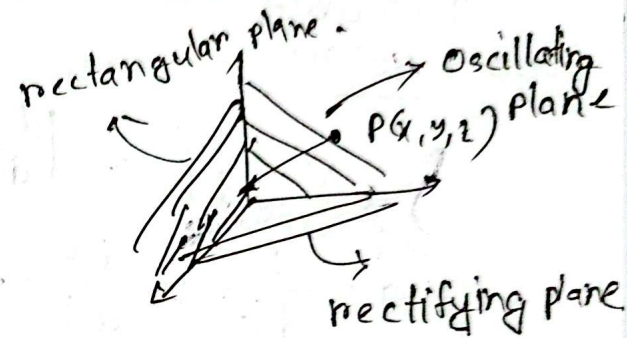
Equation of the Binormal :-

$$(\vec{r} - \vec{r}_0) \times \vec{B}_0 = 0 \quad \dots (iii)$$

Exercise-60 :

Q-24: Ex-6.1:-

Find the equations in vector and rectangular form for the (a) osculating plane, (b) normal plane, and (c) rectifying plane to the curve of Problem 22 and 23 at the point where $t = 1$



Sol n:-

Equation of osculating plane:-

$$(\vec{r} - \vec{r}_0) \cdot \vec{B} = 0$$

Equation of normal plane:-

$$(\vec{r} - \vec{r}_0) \cdot \vec{T}_0 = 0$$

Equation of rectifying plane:-

$$(\vec{r} - \vec{r}_0) \cdot \vec{N} = 0$$

Q-27: Show that the acceleration $\vec{a}$ of a particle which travels along a space curve with velocity $\vec{v}$ is given by.

$$\vec{a} = \frac{dv}{dt} \vec{T} + \frac{v^2}{\rho} \vec{N}$$

Ex-66:

Soln:

$$\vec{v} = \frac{d\vec{r}}{dt} = \left| \frac{d\vec{r}}{dt} \right| \vec{T} = v\vec{T}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv}{dt} \vec{T} + v \frac{d\vec{T}}{dt}$$

$$= \vec{T} \frac{dv}{dt} + v \frac{d\vec{T}}{ds} \cdot \frac{ds}{dt}$$

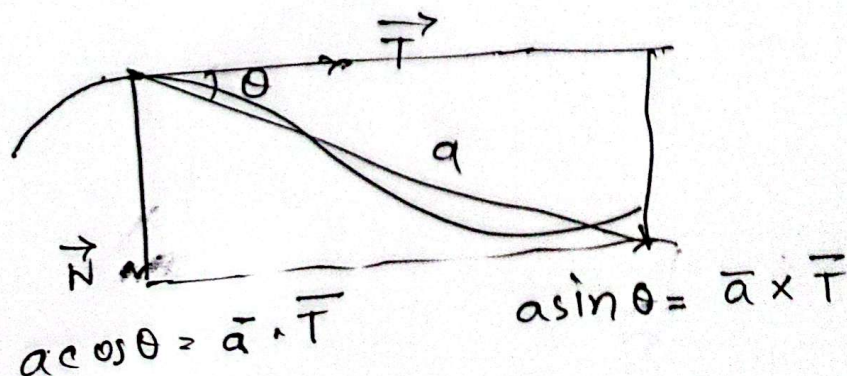
$$= \vec{T} \frac{dv}{dt} + v \cdot \kappa N \cdot \left(\frac{d\vec{r}}{dt} \right)$$

$$= \vec{T} \frac{dv}{dt} + v^2 \kappa \vec{N}$$

radius of curvature, $\rho = \frac{1}{\kappa}$

$$\vec{a} = \vec{T} \frac{dv}{dt} + \frac{v^2}{\rho} \vec{N}$$

Ex-66:



Q-25:-

(a) Show that the eq. $\vec{r} = \vec{r}(u, v)$ represents a surface.

(b) Show that the $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}$ represents a vector normal to the surface.

(c) Determine a unit normal to the following surface, where, $a > 0$:-

$$\vec{r} = a \cos u \sin v \hat{i} + a \sin u \sin v \hat{j} + a \cos v \hat{k}$$

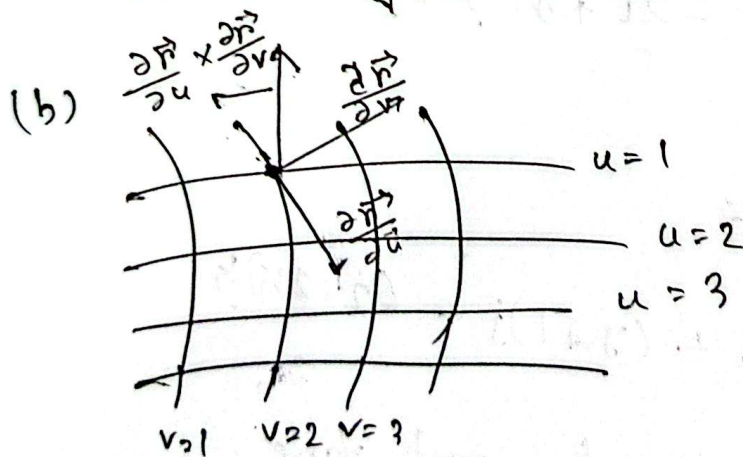
(a) Let.

$$\vec{r} = u^3 v + u^2 v^2 + \dots$$

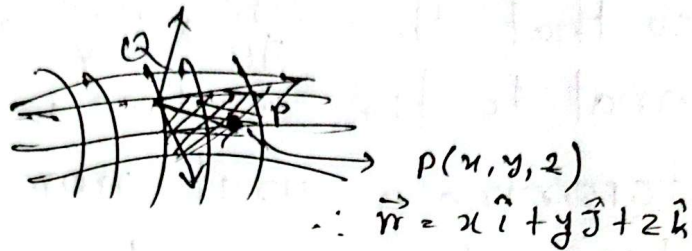
$$\text{or } r = u + v + 1$$

for first order we will get a ^{straight} plane.

and for higher order, we get curvy plane.



Q-261. Find the equation of the tangent plane and normal line to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$



Let, $x = u$, $y = v$,

$$\therefore z = u^2 + v^2$$

$$\therefore \vec{r} = u \hat{i} + v \hat{j} + z \hat{k} = (u^2 + v^2) \hat{k}$$

$$\therefore \frac{\partial \vec{r}}{\partial u} = \hat{i} + 2u \hat{k} = \hat{i} + 2\hat{k}$$

$$\frac{\partial \vec{r}}{\partial v} = \hat{j} + 2v \hat{k} = \hat{j} - 2\hat{k}$$

$$\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = -2\hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{r}_0 = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{r} - \vec{r}_0 = (x-1)\hat{i} + (y+1)\hat{j} + (z-2)\hat{k}$$

$\therefore$ equation of the tangent plane:-

$$(\vec{r} - \vec{r}_0) \cdot \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) = 0$$

Equation of the normal line:-

$$(\vec{r} - \vec{r}_0) \times \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) = 0$$